



University of Debrecen, Faculty of Informatics
Logic in Computer Science

Basic concepts of set theory

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Set and membership

The concepts of *set* and *member* are considered basic concepts without any definition. Although the *set* are usually illustrated as a collection of objects whose are called *members*.

A set can be given in many different ways:

1. by enumerating its members, e.g.:

- {spring, summer, autumn, winter} is the set of seasons,
- {0, 1, 2, 3, 4, 5, 6, 7} is the set of octal digits, but
- $\emptyset$ represents the empty set;

2. by finding a common properties of the members with a help of a condition:

- { a | a is a non-negative integer and $a < 8$ }; furthermore

3. inductively.

Membership relation

The fact that an object a is membered in a set A is written as

$$a \in A$$

where the symbol $\in$ represents the *membership relation*. To indicate that an object a is not membered in a set A , we can use the notation:

$$a \notin A$$

For example, let us denote the set of natural numbers by $\mathbb{N}$, which can be formally defined as:

$$\mathbb{N} \Rightarrow \{1, 2, 3, \dots\}$$

- 3 is a natural number, which can be expressed as $3 \in \mathbb{N}$,
- 0 is not a natural number, which can be written as $0 \notin \mathbb{N}$.

Inductively defined sets

An *inductive definition of a set* consists of declaration steps and inductive steps. By using this technique, the inductive definition of the set of natural numbers $\mathbb{N}$ can be the following:

The set of natural numbers $\mathbb{N}$ is defined inductively as follows:

1. $1 \in \mathbb{N}$, and
2. $a + 1 \in \mathbb{N}$ if $a \in \mathbb{N}$.

The first item of this definition is a simple declaration saying that an object is membered in the set. In contrast, the second item is the inductive step, showing how to generate new members from existing ones.

Basic terms and notations

Let A and B be two arbitrary sets.

- The set A is the *subset* of the set B , if all members of A are membered in B . It can be formally written as $A \subseteq B$.
- If $A \subseteq B$ and $B \subseteq A$ then A and B said to be *equal*. It can be formally written as $A = B$.
- A and B are *disjoint* if they have no element in common.

The *power set* of a set A denoted by 2^A is the set that contains all subsets of the set A .

$$\begin{aligned} 2^A &\Rightarrow \{ B \mid B \subseteq A \} \\ \text{e.g. } 2^{\{0,1\}} &= \{ \emptyset, \{0\}, \{1\}, \{0,1\} \} \end{aligned}$$

Basic operations

Let A and B be two arbitrary sets.

- *union*: $A \cup B \Leftrightarrow \{ a \mid a \in A \text{ or } a \in B \},$
- *intersection*: $A \cap B \Leftrightarrow \{ a \mid a \in A \text{ and } a \in B \},$
- *difference*: $A \setminus B \Leftrightarrow \{ a \mid a \in A \text{ and } a \notin B \},$
- *symmetrical difference*: $A \triangle B \Leftrightarrow (A \setminus B) \cup (B \setminus A).$

Let U be the universal set, that is, the set of all objects under our consideration. (A universal set is a set that contains all the elements or objects of other sets in our discussion.)

- *complement*: $\overline{A} \Leftrightarrow U \setminus A.$

Ordered pair

An *ordered pair* is a pair of objects **a** and **b** that can be formally written as:

$$\langle \mathbf{a}, \mathbf{b} \rangle$$

Two ordered pair $\langle \mathbf{a}, \mathbf{b} \rangle$ and $\langle \mathbf{c}, \mathbf{d} \rangle$ are *equal* if and only if **a** = **c** and **b** = **d**; meaning that the order in which we enumerate the objects matters.

Remark

The term “ordered pair” can be defined with the earlier discussed tools of set theory. One possible solution is the following:

$$\langle \mathbf{a}, \mathbf{b} \rangle \Rightarrow \{ \{ \{ \mathbf{a} \}, \emptyset \}, \{ \{ \mathbf{b} \} \} \}$$

This definition satisfies what is stated above regarding equality.

Cartesian product

Let A , B be two arbitrary sets. The *Cartesian product* of the sets A and B is denoted by $A \times B$ and it is defined as a set of all ordered pairs $\langle a, b \rangle$ where $a \in A$ and $b \in B$. Formally:

$$A \times B \Rightarrow \{ \langle a, b \rangle \mid a \in A \text{ and } b \in B \}$$

The Cartesian product of the sequence of sets $A_1, A_2, \dots, A_n$ ($n > 0$) can be defined by using n-tuples:

$$A_1 \times A_2 \times \dots \times A_n \Rightarrow \{ \langle a_1, a_2, \dots, a_n \rangle \mid a_i \in A_i \text{ for all } i \in \{1, 2, \dots, n\} \}$$

The A^n is an abbreviation for $\underbrace{A \times A \times \dots \times A}_{n \text{ times}}$.

Binary relations

R is called *binary relation* over the sets A and B if $R \subseteq A \times B$.

The binary relation R on a single set A is defined as a subset of $A \times A$.

The fact, that a and b are in relation R can be formally written as

$$\langle a, b \rangle \in R \quad \text{or} \quad a R b.$$

Let $R \subseteq A \times B$ be a binary relation;

- the *domain* of R is denoted by $\text{dom}(R)$ and defined as follows:

$$\text{dom}(R) \Rightarrow \{ a \mid a \in A \text{ and there is some } b \in B \text{ such that } \langle a, b \rangle \in R \},$$

- the *range* of R is denoted by $\text{rng}(R)$ and defined as follows:

$$\text{rng}(R) \Rightarrow \{ b \mid b \in B \text{ and there is some } a \in A \text{ such that } \langle a, b \rangle \in R \}.$$

Properties of binary relations

Let R be a binary relation on the set A and so $R \subseteq A \times A$.

The binary relation R is called

- *reflexive*: if $a R a$ for all $a \in A$,
- *antireflexive*: if there is no $a \in A$ such that $a R a$,
- *symmetric*: if $b R a$ for all $a R b$,
- *asymmetric*: if it is not symmetric,
- *antisymmetric*: if $a = b$, for all $a R b$ and $b R a$,
- *transitive*: if $a R c$ for all $a R b$ and $b R c$.

Properties of binary relations

Let R be a binary relation such that $R \subseteq A \times B$.

The binary relation R is called

- *injective*: if aRc and bRc then $a = b$ (for all a , b and c);
- *right total (surjective)*: if $\text{rng}(R) = B$;
- *left total*: if $\text{dom}(R) = A$;
- *bijective*: if it is both injective and surjective;
- *functional*: if aRb and aRc then $b = c$ (for all a , b and c);
- *function*: if it is functional and left total.

Notations regarding functions

If f is a function on $A \times B$ then this can be formally written as:

$$f : A \rightarrow B.$$

Set of functions

B^A represents the set of functions which has the domain A and the range B or in formal writing:

$$B^A \Leftrightarrow \{ f \mid f : A \rightarrow B \}.$$